POLS 641
Introductory Political Analysis
Class 8: Multivariate Regression

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Northern Illinois University

October 20, 2026

Agenda

- Research in action
- Course Logistics
- Warm-up Quiz
- Bivariate Review
- Multivariate Regression
- r^2

RIA

1. How would you describe the contribution?
2. What is the Unit of Analysis?
3. DV: Estimand, Estimator
4. IV: Estimand, Estimator

1. Any omitted variable bias?
2. How do they show results are ”robust”?
3. What are some weaknesses?

Course Logistics

Next Week

- Pset# 3 Not graded yet
- Pset # 4 Posted last week
- Datacamp: Only one course due Wednesday
  - Reshaping Data
  - Intermediate Regression
  - Intermediate Data Visualization
  - Reporting with Markdown
  - *Your Choice* (due Nov 25th)

Final Project - Abstract and Key Descriptive Statistics due day before Halloween

Other issues or questions?

Warm-Up Quiz

A. Interpret this regression output

```
Call:
lm(formula = nba2$ThreePerc ~ nba2$assistsgame)

Residuals:
      Min        1Q    Median        3Q       Max
-0.050857 -0.008803  0.000232  0.009252  0.046628

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.2507541  0.0191588  13.088  < 2e-16 ***
nba2$assistsgame 0.0046941  0.0008636   5.435 2.99e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.01689 on 118 degrees of freedom
Multiple R-squared:  0.2002,    Adjusted R-squared:  0.1935
F-statistic: 29.54 on 1 and 118 DF,  p-value: 2.991e-07
```

You don't need basketball knowledge to interpret

1. Can we claim causality when using multivariate regression? Why or why not?
2. What is a p-value?
3. What is the difference between an estimand and an estimator?
4. Write out our standard model: $Y =$

Warm-Up

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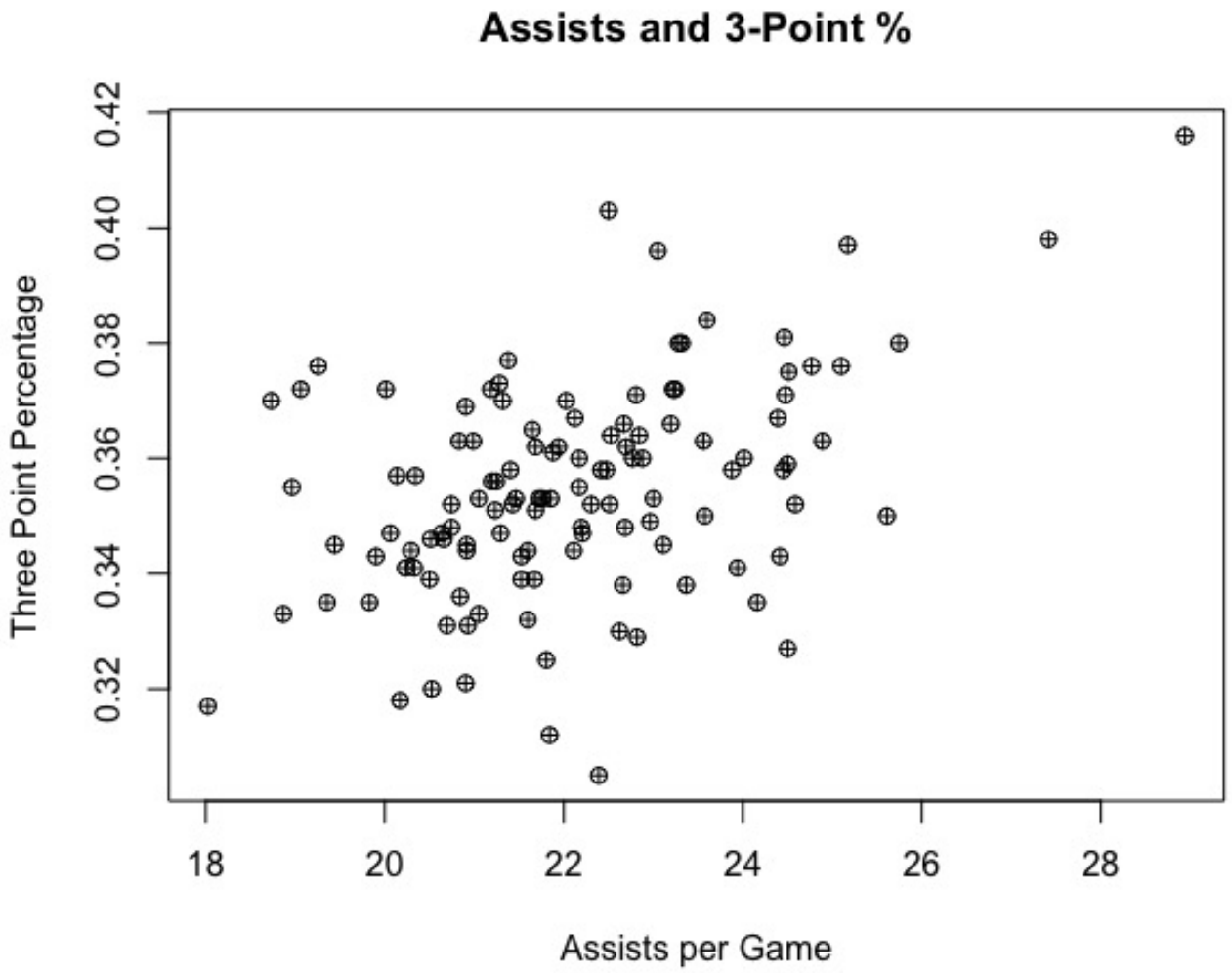
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```

1. Significance
2. Sign
3. Size
4. Substantive

Warm-Up

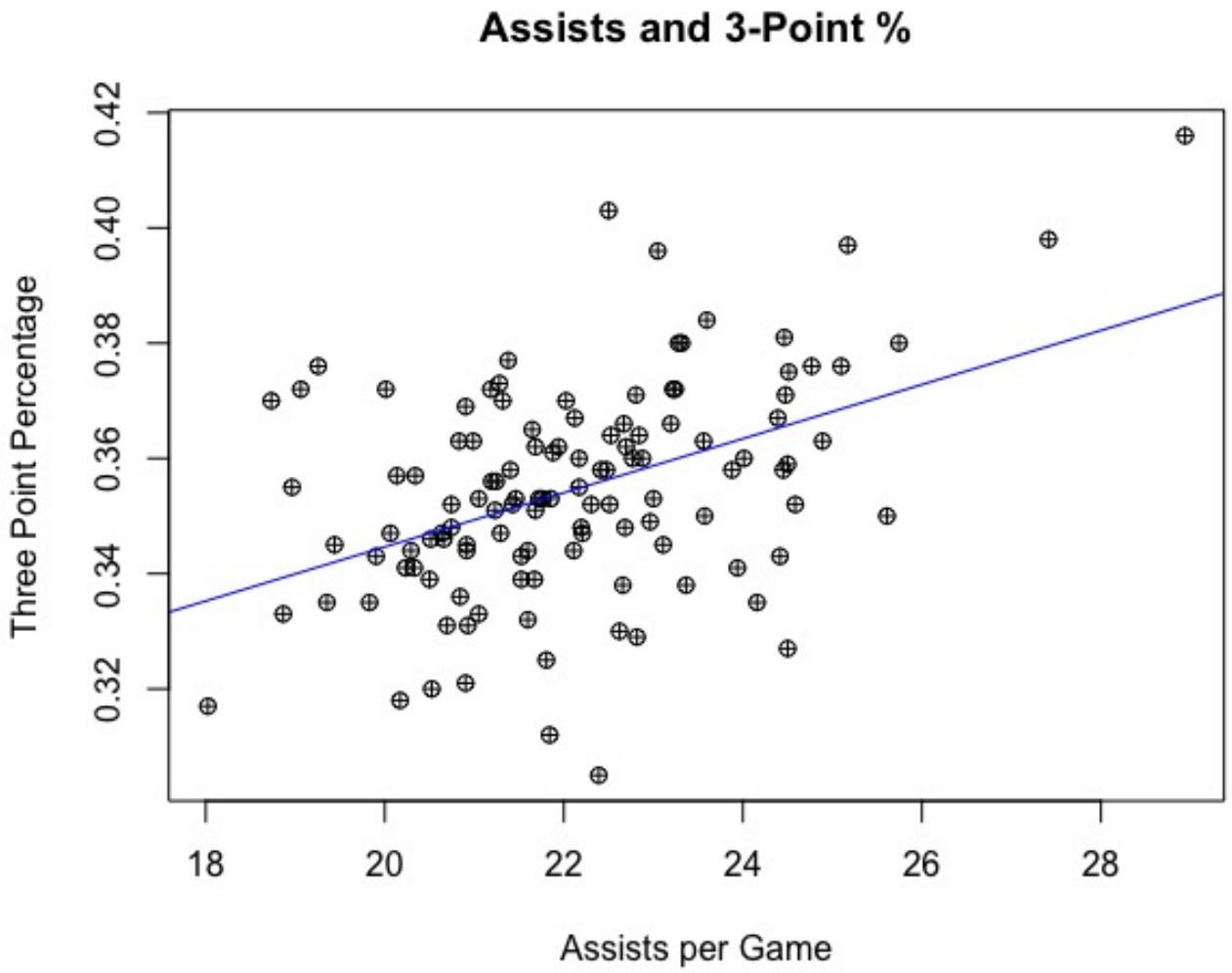
- Put into our model
- What would the visual look like
- The average team in our dataset has 22 assists per game. What is our expectation for their 3-point percentage?

Warm-Up

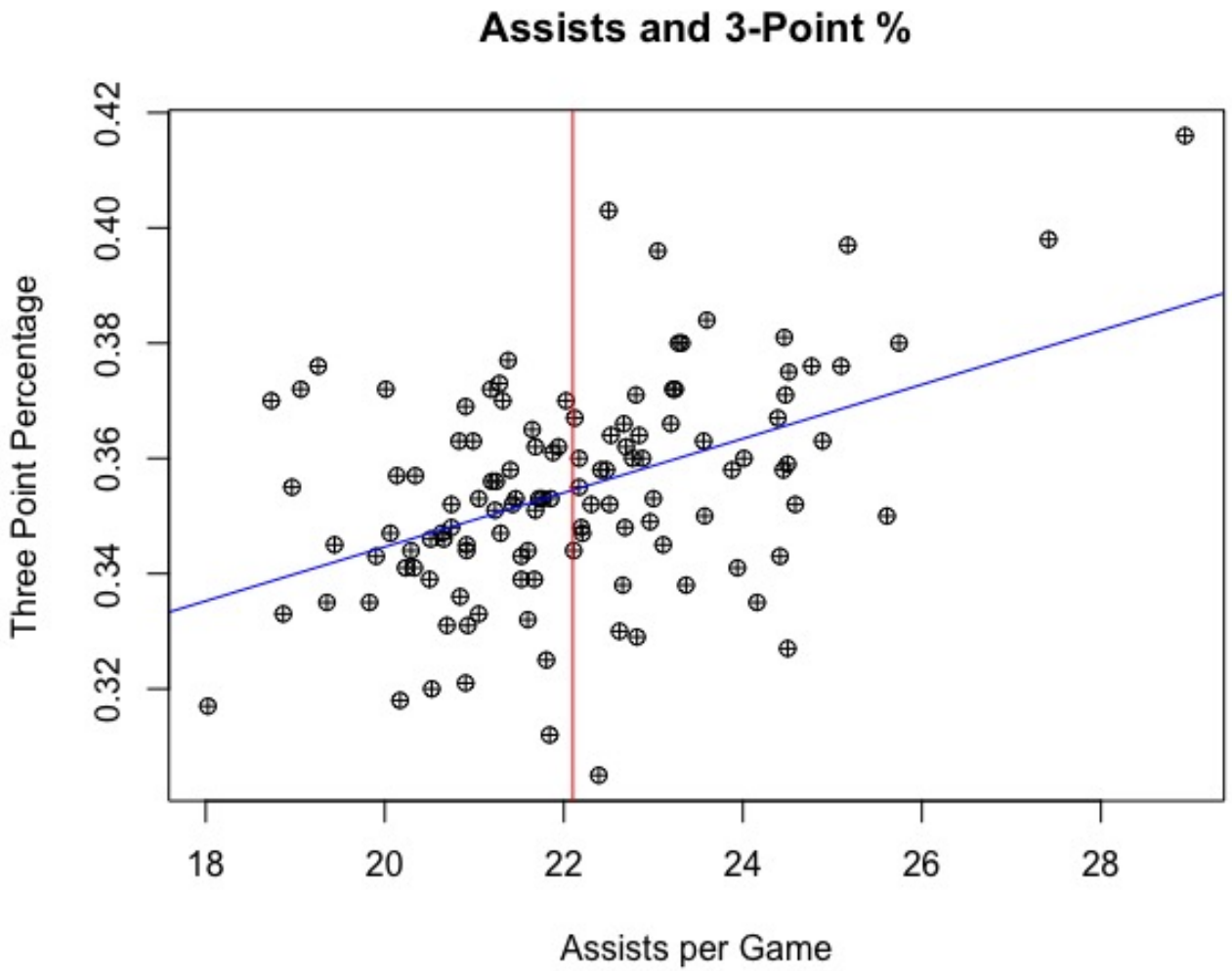

Assists and 3-Point %
Three Point Percentage
0.32
0.34
0.36
0.38
0.40
0.42
18
20
22
24
26
28
Assists per Game

1. Significance
   Positive
2. Sign
   P-value is less than .001 - less than one in a thousand chance this relationship is occurring by chance
3. Size
   For every additional assist per game the three point percentage goes up a half percent.
4. Substantive
   We expect a team generating an additional 10 assists per game to increase their 5 three point accuracy by 5 percentage points.

Warm-Up


Assists and 3-Point %
Three Point Percentage
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Assists per Game

Warm-Up


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Assists per Game

Output to Model

$$Y_i\, ThreePointPercentage = \beta_0 + \beta_1 X_i\, AssistsperGame + \epsilon_i$$

$$Y_i\, ThreePointPercentage = .251 + .00470\, AssistsperGame + \epsilon_i$$

$$Y_i\, ThreePointPercentage = .251 + .00470 * 22.1 + \epsilon_i$$

$$Y_i\, ThreePointPercentage = 35.48 + \epsilon_i$$

Questions?

Is it causal?

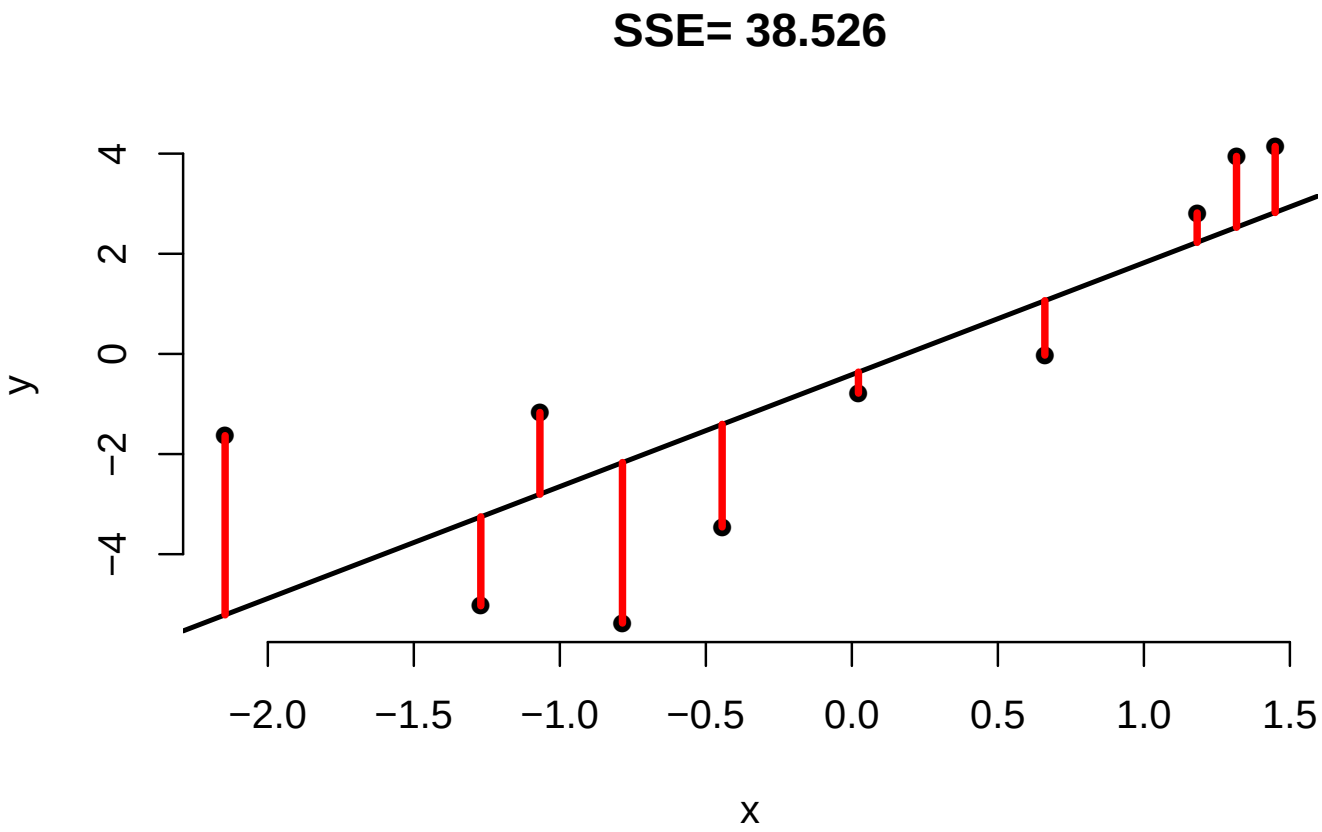
Additional Review Questions

1. Can we claim causality when using multivariate regression? Why or why not?
2. What is a p-value?
3. What is the difference between an estimand and an estimator?
4. Write out our standard model: Y =

Review: Bivariate Regression

How does OLS find the line of best fit?

Let's go back to the best fit



For this best fitting line, $\beta_0 = -.41$, $\beta_1 = cov(x, y)/var(x) = 2.24$

Bivariate Regression

What we can interpret:

1. Significance - p-value
2. Sign - coefficient positive or negative
3. Size - coefficient's value
4. Substantive - stated in real world terms

Bivariate Regression

Our core model:

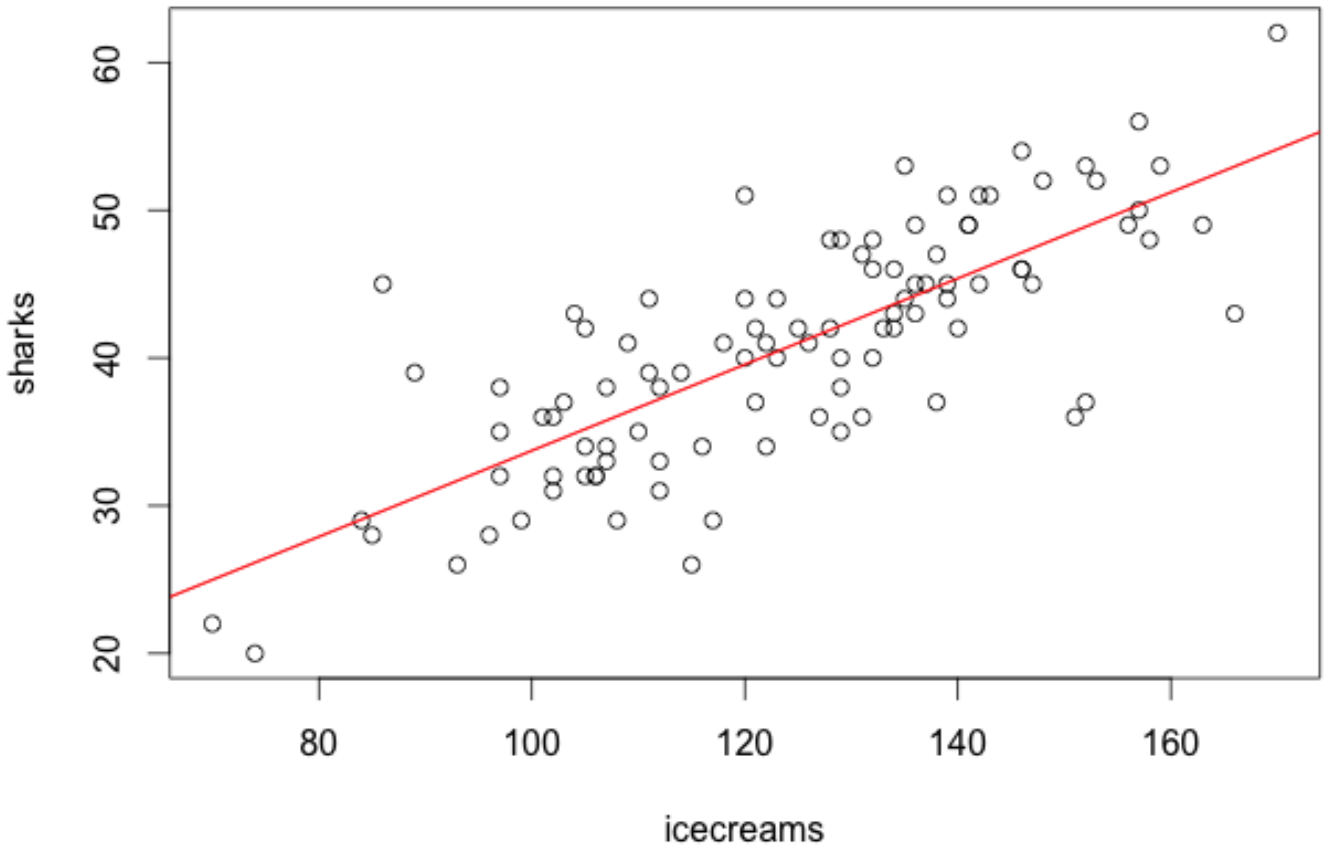
$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

or equivalently, in terms of our "best guess" $\hat{Y}_i$

$$\hat{Y}_i = \beta_0 + \beta_1 X_i$$

Remember, in order to make a causal claim, we need exogeneity
Assuming we have observational data, what might strengthen our claim about the relationship between X and Y

A Classic: Shark Attacks and Ice Cream Sales



What other factors could be impacting our dependent variable?

Multiple Predictors

But often we think there are more factors affecting our outcome variable. We need more independent/predictor variables, e.g.:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \epsilon_i$$

Why?

- to more precise with our expectation for Y_i
- to "control for" or "remove the effect of" one variable (say Z_i) when interpreting the coefficient on the other

Usually we like to call the independent variables X_1, X_2, X_3, etc. Eg.:

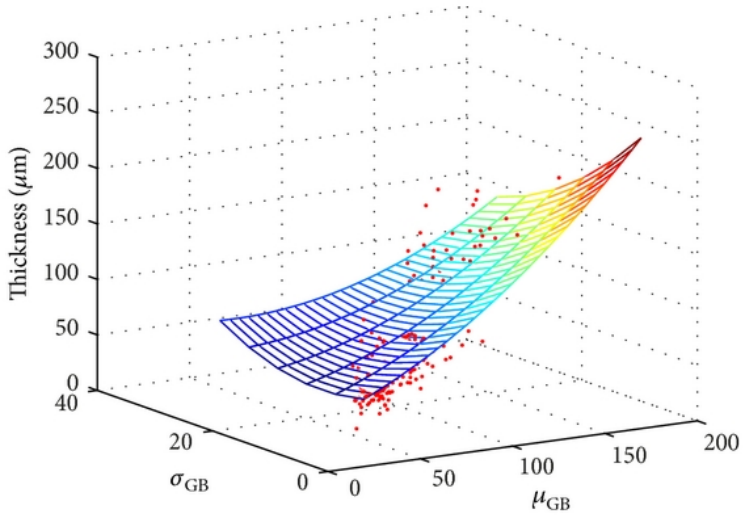
$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \epsilon_i$$

We can calculate the β's that make this model fit best in the least-squares sense, i.e.

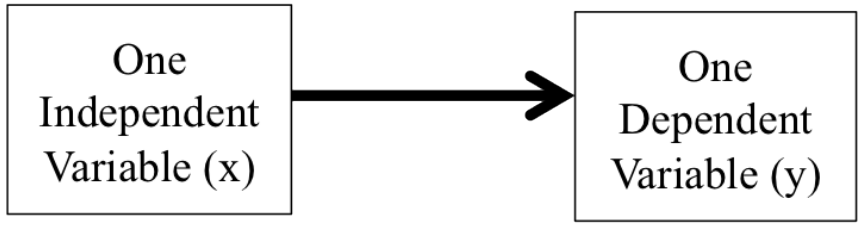
$$\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$$

though the mathematics for this is beyond this course.

Visualizing Multivariate Regression

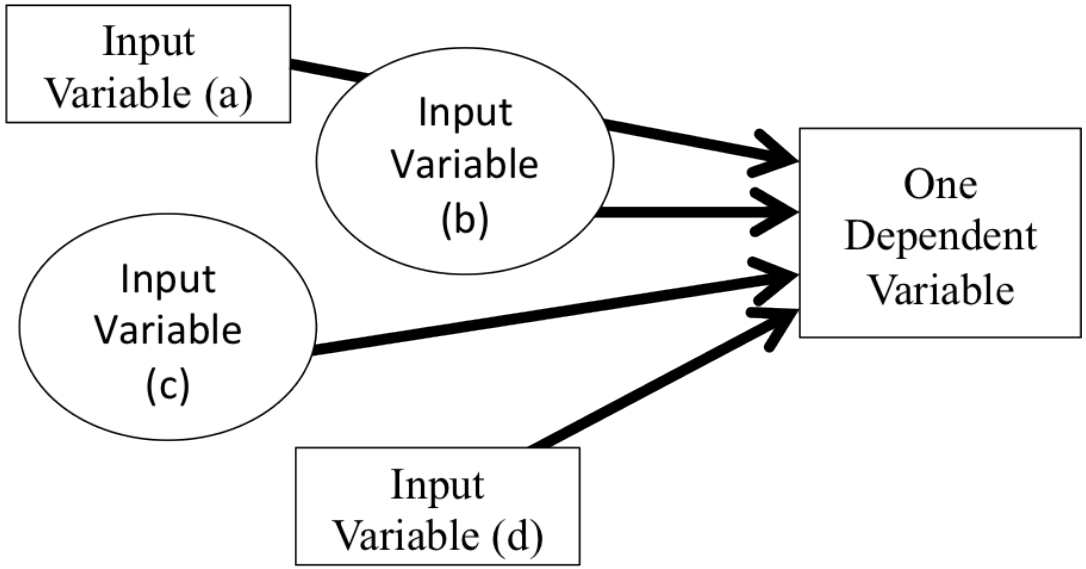


Single-Variable Regression:



Output: $y = f(x)$

Multiple Regression:



Output: $y = f(a, b, c, d)$

The Regression Interpretation and Causality

If we want to make the claim the β meaningfully estimates the causal effect of X on Y, then we need at assume that our independent variable and error term are in no way correlated. ie we want exogeneity

$$cor(X, \epsilon) = 0$$

A tricky but critical point:

- you cannot *check* whether $cor(\epsilon, X) = 0$ – it's an assumption!
- because the $\hat{\epsilon}$ you get is from the real model you have run, not the idealized causal-only model
- in the model you run, any element of ϵ correlated with X will go into falsely changing β away from it's causal value
- at the same time, this bias in β means that you are using the $\beta_1 X_1$ to remove any effect of X from Y, including what "should have" been in the ϵ.

Words

People talk about this in different ways:

- Endogeneity: we say X is "endeogenous", i.e. partially related to something that influences the outcome.
- By contrast if X comes from "outside " the system and is uncorrelated with all influences on Y_i (i.e. $cor(\epsilon, X) = 0$), we call it exogeneous.
- This is why the language of "independent variables" is a bit misleading: your "independent variable" X may be correlated with things that also influence Y.
- Omitted variable bias: the γ is bias due to not having the confounder in the model.
- So, simply, to PUT THE CONFOUNDER into the REGRESSION!

Why Multivariate Regression?

If we can include variables that would have been confounders in the regression, we take them out of the ϵ, and avoid omitted variable bias.
Suppose we are interested in the effect of X_1. So we might run:

$$growth = \beta_0 + \beta_1 X_1 + \epsilon$$

But some X_2 correlated with X_1 is also influencing Y (i.e. is a confounder. So we instead run:

$$growth = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \eta$$

- X_2 is a potential confounder, so including it in the model allows β_2 to "soak up" the effect of X_2
- had we not, our ϵ in the first model would include a function of X_2, and we said that is correlated to X_1. You can think about it like X_2 was hanging out in the error term before we put it in the model, and that was a problem.
- as a result β_1 in our first model is wrong for the causal effect: it soaks up some of the effect that should have been attributed to X_2.
- including both X_1 and X_2 in the model, we figure out what part of Y is due to X_1 and X_2, and hope nothing correlated with X is left in the new error term η

Attempting to Avoid Omitted Variable Bias

$$growth = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \eta$$

Omitted Variables Bias: Bias that results from leaving out a variable that affects the dependent variable and is correlated with the independent variable.
How can we understand this regression?

We say that:

- this "controls for" the role of X_2.
- this "soaks up the effect of X_2" so that β_1 more accurately captures the effect of X_1
- X_1 and X_2 "compete" to explain *growth*, so the one which is the better predictor will "win out"
- β_1 tells us how X_1 relates to *growth* if we could "hold X_2 constant"

Example

Does "ethnic fractionalization" (`ethfrac`) slow economic growth (`growth`)?

Warning: This phrasing makes slow a causal word, so we will be treading carefully and come back around to a safe interpretation.

Let's hypothesize that `ethfrac` does slow `growth`:

- How then would you expect `ethfrac` and `growth` to be related in your data in terms of correlation, covariance, and a regression coefficient?

Let's try the regression model,

$$growth_i = \beta_0 + \beta_1 ethfrac_i + \epsilon_i$$

- *ethfrac* is a measure of ethnic fractionalization, ranging from 0 to 1 (mean=0.45))
- *growth* is rate of growth in gdp per capita by year (mean=3.78)

Example

```
> summary(lm(rgdpgrowth~ethfrac, data=dat))
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.3121     0.2306  18.703   <2e-16 ***
ethfrac       -0.9517     0.4373  -2.176   0.0296 *
```

How do we interpret this?

- the intercept, 4.31 is our best guess of the *growth* rate when *ethfrac* $= 0$
- the coefficient on *ethfrac* is -0.952, meaning a one-unit difference in *ethfrac* corresponds to a -0.952 lower growth rate.
- do we think this slope is significantly different from zero?
  Yes. See *SE*, *t*, and *p* value.

Example

Okay, but can you make causal claim? No!!

Maybe we need to add some more variables into our regression to capture the relationship more fully.

The idea of multivariate regression is to try to "control" for or remove the effects of some potential confounders.

Let's try adding *oilproduction*, the proportion of *GDP* from oil. Why?

- perhaps by accident of history places higher in ethnic fractionalization are higher in oil production and it is this difference in oil production that is slowing growth due to its effect on the economy (the 'resource curse')
- we would call this a confounder
- so we are interested in "controlling for" *oilproduction* or asking "at a fixed level of oil production, what is the relationship between ethnic fractionalization and growth"?

Let's try running our first multivariate OLS

```
 summary(lm(rgdpgrowth~ethfrac+oilproduction_gdp, data=dat))

Call:
lm(formula = rgdpgrowth ~ ethfrac + oilproduction_gdp, data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-67.558  -3.000   0.169   3.189  66.859

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)        3.3323     0.2919  11.414   <2e-16 ***
ethfrac           -0.4823     0.5290  -0.912    0.362
oilproduction_gdp -1.3955     1.5634  -0.893    0.372
---
```

How to interpret these results?

- Now coefficient on *ethfrac* is -0.482. Iis it "significant"?
- What is the coefficient on *oilproduction*? Is it "significant"?

Got Causal?

Suppose the coefficient on *ethfrac* had remained significantly negative in our multivariate regression model after adding *oilproduction*

Then, would you feel comfortable saying *ethfrac* slows *growth*??

- "Controlling for" one potential confounder doesn't mean you have controlled for all of them! So NO, not clear to make causal claim.
- But you are probably better off having eliminated this one potential confounder or alternative explanation for the coefficient

Recall that when critiquing a causal claim, you are obligated to point out potential confounders.

Meanwhile, the researcher's goal is to include those potential confounders as control variables, when possible to rule out those concerns.

Not as good as an experiment, but we do the best we can.

REMEMBER THIS

1. Multivariate OLS is used to estimate a model with multiple independent variables.
2. Multivariate OLS fights endogeneity by pulling variables from the error term into the estimated equation.
3. As with bivariate OLS, the multivariate OLS estimation process selects $\hat{\beta}$'s in a way that minimizes the sum of squared residuals.

What variables?

What variables?

- Literature
- Theory
- Of course limited to data availability and subject to all caveats on data quality and measurment validity

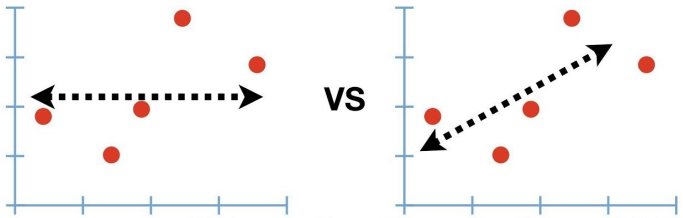
- Avoid the kitchen sink regression. Balance parsimony and power concerns with comprehensiveness

Brainstorm

What variables should be included in a model predicting the likelihood of an individual voting?

Understanding the "Variance Explained" or R^2

The "spread" of points around the regression line should be smaller than the spread of points around the mean-line.



- if we average up the squared distances around the mean line we get the variance of Y.
- if we average up the squared distances around the regression line we get mean squared error (MSE) for the regression
- we were trying to minimize this form of prediction error in our choice of β. You could say $\frac{MSE}{Var(Y)}$ is the proportion of variance of Y that remains unexplained by the model. Thus $1 - \frac{MSE}{Var(Y)}$ is the proportion of variance "explained'
- this is R^2 statistic at the bottom of the table

- R^2 is interpreted as the proportion of variance explained by the model as whole.
- R^2 is equal to the square of the correlation between the fitted and observed values of the dependent variable.
- R^2 **can also be interpreted as the proportion of variance explained.**
- A highR^2 is neither necessary nor sufficient for an analysis to be useful. A good model can have a low r^2 and a bad model can have a high r^2

- R^2 is our primary means of making comparisons between models

adjusted r^2

- Adding additional variables necessarily increases the r^2
- Adjusted r^2 takes the number of variables into account(using math beyond our scope)
- Interpreted the same way as normal r^2. Usually what is reported when presenting multivariate models.

Review of key points on OLS so far...

- We can estimate the relationship between X & Y, with two key coefficients: $\hat{\beta}_1$ & $\hat{\beta}_0$
- Due to sampling variation, our estimates also have a standard error
- Our model creates fitted values, $\hat{Y}$
- Our model has residuals, the part of Y that X(s) does not explain, $\hat{\epsilon}_i$
- Including additional ”control" variables decreases omitted variable bias
- Relatedly, we can estimate our model’s ‘goodness of fit’: R^2
- Just because we see a significant relationship in our regression model, doesn’t mean its causal

Review: Interpretation

For Each Independent Variable:

1. Significance
2. Sign
3. Size
4. Substantive

(Which variables and to what extent we interpret them is dependent on our goals for the paper) For the model as a whole:

1. R^2 Proportion of variance explained

Key Terms

- Omitted Variable Bias
- Multivariate Regression
- Ceteris Paribus
- Confounder
- Control Variable
- R^2
- Adjusted R^2
- Goodness of fit

We will talk more about issues with multivariate regression(multicollinearity, autocorrelation, heteroskedasticity, etc in the coming weeks)